

**Fundamentals of GIS:**

Introduction, data models (raster/vector), spatial analysis, geodatabases, topology,

**Unit- 2.0:** 5 hrs

**Remote Sensing Principles & Applications:**

EM spectrum, sensors, platforms, image processing, classification, corrections.

**Unit- 3.0:** 5 hrs

**Cartography and Map Projections:**

Map design, types of maps, coordinate systems, datums, digital mapping, visualization.

**Unit- 4.0:** 4 hrs

**GNSS (GPS) & Surveying**

GPS, GLONASS, Galileo, data collection methods, accuracy, error sources, modern surveying techniques.

**Unit- 5.0:** 6 hrs

**Geospatial Programming & Databases**

Python programming (ArcPy), spatial SQL, database management systems, automation, web GIS basics.

**Unit- 6.0:** 4 hrs

**Applications:**

Case studies in urban planning, disaster management, agriculture, forestry, EIA.

**Text /Reference:**

1. Longley, P. A., Goodchild, M. F., Maguire, D. J., & Rhind, D. W. (Geographic Information Systems and Science). John Wiley & Sons.
2. Lillesand, T. M., Kiefer, R. W., & Chipman, J. W. (Remote Sensing and Image Interpretation). John Wiley & Sons.
3. Robinson, A. H., Morrison, J. L., Muehrcke, P. C., Kimerling, A. J., & Guptill, S. C. (Elements of Cartography). John Wiley & Sons.

**Unit- 1.0: Basic Concepts and Definitions****7 hrs**

Distinction between a fluid and a solid; Density, Specific weight, Specific gravity, Kinematic and dynamic viscosity; variation of viscosity with temperature, Newton law of viscosity;